EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Changing a by-value int parameter always changes the caller's int.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

# byVal, byRef and modular programming

Name: _____ Class: _____ Date: _____

**1. What does return do in a C# function?**

**2. Can a method read a local variable declared inside another method?**

**3. What is a[0] for int[] a={9,6,3}?**

**4. If Double(n) returns 2*n, find Double(Double(3)).**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Finishes the invocation and returns a result to its caller.

**2. Model answer**

Not directly. It is outside the variable's scope.

**3. Model answer**

9.

**4. Model answer**

Double(3)=6; Double(6)=12.

**Vocabulary — know these terms**

by value — The subroutine receives a copy of the value.

by reference — The parameter refers to the caller's variable.

modularity — Divide a program into smaller testable subroutines.

ref — C# keyword enabling reference parameter passing.

side effect — A change to external program state.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 3**

Explain when returning a result is preferable to changing a ref parameter.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 1**

A method takes int number by value and adds 2. Caller x=5. What is x after the call?

**Question 2**

Write a C# procedure that doubles an integer passed by reference.





NEW LEARNING

# Read, annotate and explain

**CALLER**

The caller declares marks = 10.

**ALIAS**

ref makes score refer to the caller's marks rather than an independent copy.

**AFTER**

Changing score therefore changes marks, which becomes 15.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**A function receives a copy of x=7 and adds 4. What is x afterwards?**

**Predict first**

A new parameter value receives a copy of the integer 7.

**Model the result**

value becomes 11 inside AddFour, but x remains 7 after the call.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve